

RaceDay – API Endpoint Plan

Part 1: System Planning and Database — endpoint specification for the RaceDay platform. This plan must be matched closely by the implemented API in Part 2.

1. Authentication

Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register a new user as an Organiser or a Participant.	Public	{ FullName, Email, Password, Role }	201 Created – new user object (Email, FullName, Role, UserID); password never returned.
POST	/api/auth/login	Authenticate a user and issue an access token.	Public	{ Email, Password }	200 OK – { token, UserID, FullName, Role }. 401 Unauthorized on bad credentials.

2. User Profile

Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Get the logged-in user's own profile.	Authenticated	None	200 OK – user profile object (UserID, FullName, Email, Role, PhoneNumber, CreatedAt).
PUT	/api/users/me	Update the logged-in user's profile details.	Authenticated	{ FullName, PhoneNumber }	200 OK – updated profile object.
GET	/api/users/me/results	Get the logged-in participant's personal performance history.	Participant	None	200 OK – array of past results (EventName, CategoryName, FinishTime, Position).

3. Events

Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	List all upcoming events, optionally filtered by type or location.	Public	None	200 OK – array of event summaries.
GET	/api/events/{id}	Get full details for a single event, including its categories.	Public	None	200 OK – event object with nested categories. 404 Not Found if it doesn't exist.
POST	/api/events	Create a new event.	Organiser	{ EventName, EventType, EventDate, Location, Description }	201 Created – new event object.
PUT	/api/events/{id}	Update an existing event owned by the logged-in organiser.	Organiser (owner)	{ EventName?, EventDate?, Location?, Description? }	200 OK – updated event object. 403 Forbidden if not the owner.
DELETE	/api/events/{id}	Cancel/remove an event owned by the logged-in organiser.	Organiser (owner)	None	204 No Content. 403 Forbidden if not the owner.
GET	/api/events/{id}/weather	Get a live weather forecast for the event's date and location, to help participants prepare.	Public	None	200 OK – { temperature, conditions, windSpeed, forecastDate } from external weather API.

4. Categories

Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	List all categories (distances) for a specific event.	Public	None	200 OK – array of category objects.
POST	/api/events/{eventId}/categories	Add a new category to an event.	Organiser (owner)	{ CategoryName, DistanceKM, EntryFee, MaxParticipants }	201 Created – new category object.
PUT	/api/categories/{id}	Update a category's details.	Organiser (owner)	{ CategoryName?, DistanceKM?, EntryFee?, MaxParticipants? }	200 OK – updated category object.
DELETE	/api/categories/{id}	Remove a category from an event.	Organiser (owner)	None	204 No Content. 409 Conflict if participants are already enrolled.

5. Event Enrolments

Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Enrol the logged-in participant into a category of an event.	Participant	{ CategoryID }	201 Created – enrolment object including generated RaceNumber and PaymentStatus.
GET	/api/users/me/enrolments	List all events the logged-in participant is enrolled in.	Participant	None	200 OK – array of enrolment objects with event and category details.
GET	/api/events/{eventId}/enrolments	List all participants enrolled in an event (for the organiser).	Organiser (owner)	None	200 OK – array of enrolments with participant details.
DELETE	/api/enrolments/{id}	Cancel the logged-in participant's own enrolment.	Participant (owner)	None	204 No Content. 403 Forbidden if not the owning participant.

6. Results

Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{id}/results	Capture a finishing result for a participant's enrolment.	Organiser (owner)	{ FinishTime, Position, Status }	201 Created – new result object.
GET	/api/events/{eventId}/results	Get the full results list for an event (leaderboard).	Public	None	200 OK – array of results ordered by Position.
GET	/api/results/{id}	Get a single result record by ID.	Authenticated	None	200 OK – result object. 404 Not Found if it doesn't exist.

Note: routes marked "(owner)" additionally require the authenticated user's UserID to match the OrganiserID on the parent Event, or the ParticipantID on the parent Enrolment, enforced server-side after JWT authentication.